



1. Interior of the Earth

IN THIS CHAPTER, YOU WILL...

- Learn about the different layers of the earth.
- Learn about the types of rocks found on the earth's surface and how they are formed.

CONNECTIONS TO OUR LIFE

- The land we walk on is made of rocks as well as minerals which have many uses.
- Some rocks look different from others because of the way they were formed.

BIG IDEA

The earth is made up of different layers. The rocks and minerals that are found in these layers are important to us in many ways.

LET'S BEGIN

Have you ever dug a deep hole while playing in the park? Imagine you could dig deeper for several kilometres. What do you think you would find? In Class 5, we learned that there are several layers below the surface of the earth. Let us do an activity to recall what we know about them.

ACTIVITY 1

- Draw and label the layers of the earth in your notebook.
- Write any one feature of each layer.

THINK

In reality, it is not possible to dig a hole through the earth. Then, how did scientists discover the structure of the earth? Search the internet to learn more about this.

In this chapter, we will learn about the features of the different layers of the earth. We will also learn about the types of rocks that form the outermost layer of the earth.

1.1 LAYERS OF THE EARTH

We learned that the earth is the only planet known to support life. But how was our planet created? How were its layers formed? Let us find out.

Over 4.6 billion years ago, a giant cloud of gas and dust formed in outer space after the Big Bang. This cloud of gas and dust, called a 'nebula', was held together by gravity. Later, this nebula collapsed and formed the sun, the planets, and other celestial bodies that we see in the solar system.

At its beginning, the earth was not a liveable planet. It was an extremely hot, big ball of molten rock. As it cooled down over millions of years, different layers were formed. The heavier materials sank towards the centre, forming the core, while the lighter materials floated above, forming the crust.

Structure of the Earth



Have you ever sliced an onion? The different layers of the onion resemble the structure of the earth. Look at Fig. 1.2 to see what each layer is called.

- Crust:** It is the outermost layer of the earth.
- Mantle:** It is the second layer and lies below the crust.

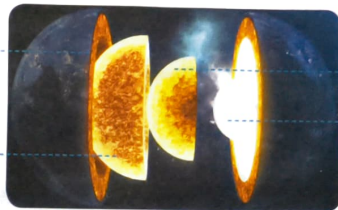


Fig. 1.2: Layers of the earth

- Outer Core:** It is the third layer and lies below the mantle.
- Inner Core:** It is the innermost layer of the earth.



Fig. 1.1: A nebula

KEYWORDS

gravity: a strong, invisible force that pulls objects towards each other
molten: heated to a very high temperature so that it becomes liquid

1

Crust

- It is the **thinnest layer** of the earth and is made of solid rocks.
- It is divided into continental crust (land) and oceanic crust (ocean floor).
- The continental crust is mainly made up of silica and aluminium, also called 'sial'. The oceanic crust is mainly made up of silica and magnesium, also called 'sima'.
- The **thickness** of the crust varies from 5 to 8 kilometres under the oceans and 35 to 70 kilometres under the continents.
- Its **temperature** ranges between 0°C and 700°C .

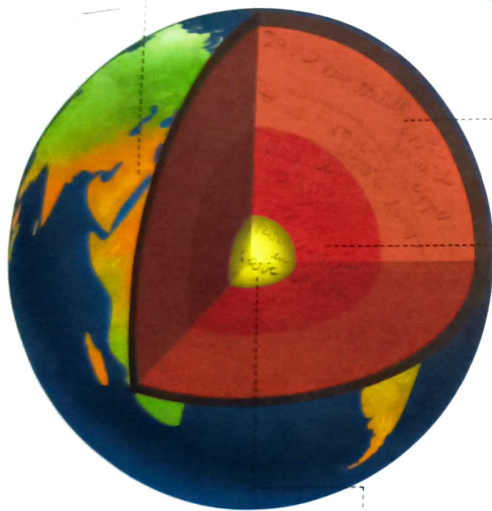


Fig. 1.3 Structure of the earth

2

Mantle

- It is the **thickest layer** of the earth.
- Its thickness is about **2,900 kilometres**.
- It is divided into the upper mantle and the lower mantle.
- It is made up of semi-liquid or molten rocks. The rocks largely contain silicon, iron, and magnesium.
- Its temperature ranges between 870°C and 4400°C .

3

Outer Core

- It is the **liquid part** of the core.
- It is about **2,300 kilometres** thick.
- It is mainly made up of nickel and iron (**ferrum**) and is also referred to as 'nife'.
- Its temperature ranges between 4400°C and 6100°C .

4

Inner Core

- This layer is about **1,200 kilometres** thick.
- It has the same composition as the outer core, but it is solid **because of high pressure**.
- Its **temperature** is as high as 6000°C .

DID YOU KNOW?

The hottest part of the earth's core, where the temperature reaches 6000°C , is as hot as the surface of the Sun.

ACTIVITY 2

Make a well-labelled model of the earth's layers using clay of different colours or other materials.

1.2 ROCKS AND MINERALS

If you dig deep into the soil in your playground, you will eventually hit something hard. What do you think it is? It is a rock!

Rocks are naturally occurring substances that form a part of the earth's outermost layer. These rocks make up many things around us — from tall mountains in the Himalayas to the soil in our gardens. Rocks are made up of one or more minerals. **Minerals** are naturally occurring substances that contain one or more elements. Different minerals have different chemical and physical properties. We will learn more about minerals later in the chapter.



Fig. 1.4 A layer of rocks under the soil

Most **rocks** may **appear** the same to us, but they vary in **shape, size, colour, and texture**. These features are determined by the way rocks form over time. Based on formation, rocks are classified into three types — **igneous rocks**, **sedimentary rocks**, and **metamorphic rocks**. Let us learn more about each of them.

Igneous Rocks



The interior of the earth is made of a very hot molten material called **magma**. When this molten material cools and **solidifies** inside or on the surface of the earth, the rocks formed are called **igneous rocks**. Let us see how this happens.

- Due to high heat, rocks melt inside the earth. They mix with gases and other rocks, forming magma.
- During volcanic eruptions, magma is pushed from the mantle towards the surface of the earth.
- The magma that flows out onto the surface of the earth due to volcanic eruptions is called lava.
- On the surface of the earth, the lava cools down, hardens, and forms igneous rocks.

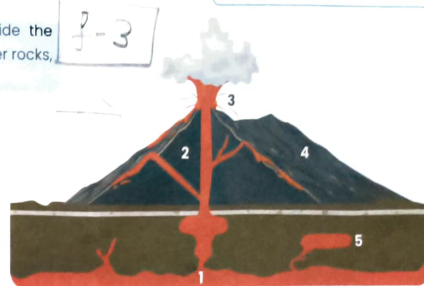


Fig. 1.5 Formation of igneous rocks

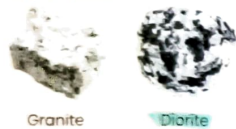
- Sometimes, instead of flowing out onto the surface of the earth, magma can also solidify somewhere within the crust, forming igneous rocks.

Igneous rocks are also called primary rocks since these were the first rocks to be formed. All other types of rocks are directly or indirectly formed from igneous rocks.

Igneous rocks are classified into two types: intrusive igneous rocks and extrusive igneous rocks.

Intrusive Igneous Rocks

These rocks are formed when magma cools down and solidifies inside the earth's crust. Since the earth is warmer inside the rate at which the magma cools down is slower. Therefore, these rocks develop large crystals.



Granite

Diorite

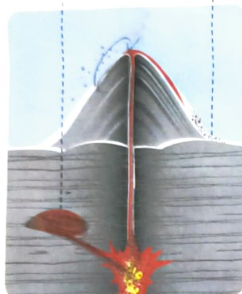


Fig. 16 Formation of intrusive and extrusive igneous rocks

Extrusive Igneous Rocks

These rocks are formed when magma pours out of the earth's surface as lava and solidifies. Since the earth is cooler on the outside, the rate at which lava cools down is faster. Therefore, these rocks have a grainy texture with small crystals.



Basalt

Andesite

Igneous rocks are used for a variety of purposes. For instance, **granite** is used for making kitchen countertops and floors in houses. Similarly, **basalt** is used for making pavements, statues, and monuments.

Sedimentary Rocks

Due to the action of wind, water, and ice, the existing rocks on the earth break down into small particles called sediments. When these sediments compress under pressure from the layers above, they form layered rocks. These rocks are called **sedimentary rocks**. Let us see how this happens.

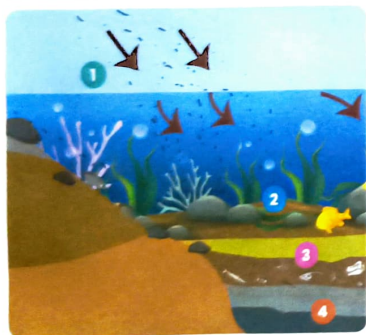


Fig. 17 Formation of sedimentary rocks

Some examples of sedimentary rocks are shown below.



Fig. 18 Examples of sedimentary rocks

KEYWORDS

fossils remains of dead plants and animals
cyclic repeated many times

Metamorphic Rocks

Sedimentary rocks often contain fossils. Over time, these fossils turn into mineral fuels such as coal, petroleum, and natural gas. Apart from containing sources of energy, sedimentary rocks are also used for making cement, tiles, bricks, and pottery.

When igneous or sedimentary rocks are exposed to extreme heat or pressure, their physical and chemical properties change. This results in the formation of **metamorphic rocks**. The word 'metamorphic' comes from two Greek words, *meta*, which means 'change', and *morphe*, which means 'form'. Let us learn how metamorphic rocks are formed.

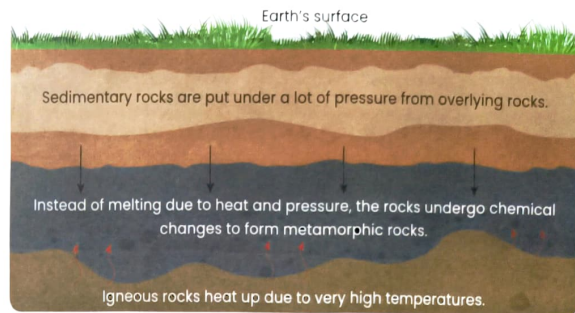


Fig. 19 Formation of metamorphic rocks

A few examples of the transformation of metamorphic rocks include the changing of granite (igneous rock) into gneiss, limestone (sedimentary rock) into marble, and clay (sedimentary rock) into slate. Different metamorphic rocks are used for different purposes. Some of these are as follows:

- Gneiss is used for making floors, tiles for walls, and kitchen countertops.
- Marble is used for making statues and floors.
- Slate is used for making roof tiles and blackboards.

CONNECT TO LIFE

Observe your surroundings and list three things that are made of rocks. Work in pairs to identify the type of rock used for making each thing.

Rock Cycle

We now know that there are three main types of rocks. **Rocks constantly transform from one type to another under certain conditions in a cyclic manner. This process is called the rock cycle.**

You must be wondering how these changes occur and why we never notice them in action. This transformation happens due to the effects of weathering (the breaking down of rocks), deposition (the laying down of sediments), cooling, heating, and pressure. As these processes are very slow and take millions of years, it is hard for us to notice them. Let us now learn about the rock cycle in detail.



Fig. 1.10 Rock cycle

- Sometimes, igneous rocks can directly become metamorphic rocks under heat and pressure.
- Metamorphic rocks can become sedimentary rocks when they are broken down by wind, water, or ice into sediments.

Minerals

You would have come across words like calcium, iron, sodium, and zinc while watching advertisements for healthy food. How are these related to rocks? Actually, these are minerals.

Minerals are the substances that make up rocks. Rocks can be made up of two or more minerals. For example, granite is made up of three minerals: quartz, feldspar, and mica.

Minerals are classified into two types — metallic minerals and non-metallic minerals.

- Metallic minerals** are minerals that contain metals. Iron, copper, tin, and manganese are examples of metallic minerals.
- Non-metallic minerals** are minerals that do not contain metals. Potash, gypsum, limestone, and mica are examples of non-metallic minerals.

Like rocks, minerals are very useful for living beings.

Minerals such as calcium, zinc, and magnesium help keep our bodies healthy.

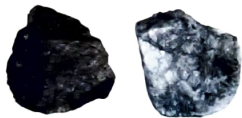


A mineral called fluoride is added to toothpaste to make teeth stronger.



Granite = Quartz + Feldspar + Mica

Fig. 1.11: The mineral breakdown of granite



Iron

Gypsum

Fig. 1.12: Types of minerals

- Igneous rocks are formed when magma or lava cools down.
- Wind, water, and ice break down igneous rocks into sediments.
- The sediments get moved and deposited in layers to form sedimentary rocks.
- Sedimentary rocks change into metamorphic rocks when put under heat and pressure.
- Under extreme heat and pressure, metamorphic rocks melt to form magma.



to learn about the rock cycle.

Diamond, the hardest mineral, is used for making jewellery.



Minerals such as limestone are used in the construction of buildings.



Plants use minerals such as potassium and magnesium to grow and build their cells.



Copper is used for making wires, coins, and pipes.

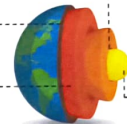


LET'S REVISE

The interior of the earth is divided into different layers.

Crust: It is the thinnest layer of the earth and is made up of silica, aluminium, and magnesium.

Mantle: It is the thickest layer of the earth and is made up of silicon, magnesium, and iron.



Outer Core: It is the liquid part of the core and is made up of nickel and iron.

Inner Core: It is the solid part of the core and is made up of iron.

The earth's crust, the layer we live on, is composed of different rocks and minerals. **Rocks** are naturally occurring substances that form a part of the earth's outermost layer. These rocks are classified into three types based on their formation.

1 Igneous rocks are formed when magma (or lava) cools down and solidifies inside or on the surface of the earth.



2 Sedimentary rocks are formed in layers when sediments at the bottom are compressed due to pressure from the layers above.



3 Metamorphic rocks are formed when igneous and sedimentary rocks are exposed to extreme heat and pressure, changing their physical and chemical properties.



All types of rocks constantly transform from one type to another in a cyclic manner. This process is called the rock cycle.

On the other hand, **minerals** are naturally occurring substances with specific chemical and physical properties and contain one or more elements. Minerals make up rocks. They are further classified as metallic and non-metallic minerals.

EXERCISES

A. Fill in the blanks.

- The earth's crust is divided into the continental crust and the oceanic crust.
- The outer core is mainly made up of nickel and iron.

- Igneous rocks are also called intrusive rocks.
- ~~Extrusive~~ rocks are formed when magma cools down and solidifies inside the earth.
- Extrusive igneous rocks have a grainy texture with small crystals.
- The weight of the sediments compresses the bottom layers of the sediments to form metamorphic rocks.
- The cyclic process in which rocks constantly transform from one type to another type under certain conditions is called the rock cycle.
- The minerals that do not contain metals are called non-metallic minerals.

B. Choose the correct answer.

- Which is the thinnest layer of the earth?
(a) Mantle (b) Crust (c) Inner core (d) Outer core
- Which of the following is not an example of metamorphic rocks?
(a) Marble (b) Slate (c) Gneiss (d) Shale
- When do metamorphic rocks become sedimentary rocks?
(a) When sedimentary rocks are broken down by wind, water, or ice into sediments
(b) When igneous rocks are under heat and pressure
(c) When metamorphic rocks are broken down by wind, water, or ice into sediments
(d) When magma or lava cools down
- Which of the following is not an example of a metallic mineral?
(a) Iron (b) Copper (c) Tin (d) Potash
- Which of the following minerals are used by plants to grow and build their cells?
(a) Potassium and granite (b) Diamond and magnesium
(c) Potassium and mica (d) Feldspar and mica

C. Write True or False. Rewrite the false statements correctly in your notebook.

- The inner core is the liquid part of the core. False
- Sedimentary rocks often contain fossils. True
- Granite is an example of an igneous rock. True
- Different minerals have the same chemical and physical properties. False
- Diamond, the hardest mineral, is used for making jewellery. True

D. Draw the table given below in your notebook to describe how each type of rock is formed. Include one example of each type of rock.

Rock Type	How is it formed?	Example
Igneous rock		
Sedimentary rock		
Metamorphic rock		

E. Answer the following questions in brief:

- List any three features of the mantle.
- Why is the inner core solid?

- What are rocks? List three uses of rocks.
- Why are igneous rocks called primary rocks?
- What are minerals?

F. Answer the following questions in detail:

- Differentiate between the following
(a) The inner core and the outer core
(b) Intrusive igneous rocks and extrusive igneous rocks
- Describe the structure of the earth.
- Explain the formation of igneous rocks.
- Explain the rock cycle with the help of a labelled diagram.
- Define the different types of minerals.

G. HOTS – Think and answer.

- Do you think humans can reach the core of the earth? Why or why not?
- Why are sedimentary rocks important for humans?

H. Competency-based questions

- Look at the picture and answer the following questions:
(a) It is a type of igneous rock. It has a grainy texture and small crystals. Identify the type of rock.
(b) Explain how this texture is formed.
- Ravi wants to build a house. What types of rocks can he use, and for what purposes?



REFLECTIONS

- What are two new things that you learned in this chapter?
Igneous rocks also change to sedimentary and metamorphic rocks.
- Which topic do you feel most confident about?
Concept of erosion, the soil etc.

1. 2) Extrusive and intrusive rocks are formed from magma.



2. Our Changing Earth

IN THIS CHAPTER, YOU WILL ...

- Learn about forces acting below the surface of the earth and the impact that they have.
- Understand what volcanoes are, **how they are formed**, and their different types.
- Understand why earthquakes occur and how we can stay safe when they occur.
- Learn about the formation of different landforms on the earth.

CONNECTIONS TO OUR LIFE

- The various landforms that we see around us, such as mountains and valleys, have developed over time.
- Sudden events, such as earthquakes and volcanoes, are caused by changes within the earth.

BIG IDEA

The earth and all its landforms are constantly changing due to various **forces** that act on and below the earth's surface.

LET'S BEGIN

In earlier classes, we learned that the earth has four domains that help make life possible. Do you remember what they are? These four interconnected domains are the lithosphere (land), hydrosphere (water), atmosphere (air), and biosphere (life).

We learned that the **lithosphere** has different landforms such as mountains, plateaus, and plains. Apart from these, there are many other landforms on the earth's surface. Did you know that there was a time when many of these landforms did not exist? That's right!

The **lithosphere** is constantly changing. Many of these changes happen so slowly that we do not even notice them.

But why do these changes happen? To understand this, let us see why any change happens. Why does a door open or close? It does so when we apply a **force** to it. Similarly, changes happen on the earth because of some forces acting on it. Some of these forces are familiar to us, such as the wind that blows leaves off a tree or rivers that flow across different parts of the land. Some forces, however, are not like the ones we can feel directly. They occur deep inside the layers of the earth. Depending on where these forces occur, we can divide them into the following two categories:

Internal or endogenic forces

endo = inside

Forces that occur below the surface of the earth



External or exogenic forces

exo = outside

Forces that occur above the surface of the earth

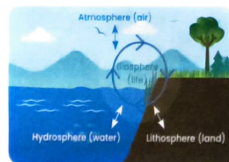


Fig 2.1: Domains of the earth

ACTIVITY 1

Draw two landforms that you have learned about. Then, discuss with your partner how you think they were formed.

KEYWORDS

force: the push or pull experienced by an object

THINK

Think about some landforms you have seen around you that might have been affected by forces from inside or outside the earth.

2.1 ENDOGENIC FORCES

Endogenic forces or internal forces are forces that act below the surface of the earth.

But what exactly is this force or movement taking place below the surface of the earth? To understand this, we need to learn about a theory known as plate tectonics.

Plate Tectonics



We know that the land on which we live is the lithosphere. But did you know that the lithosphere is not one complete piece of land? It is actually made up of several big and small pieces that fit together like a jigsaw puzzle. These individual pieces of the lithosphere are known as **lithospheric plates** or **tectonic plates**. There are seven large plates known as major plates and several smaller or minor tectonic plates. The map on the next page shows the different tectonic plates.



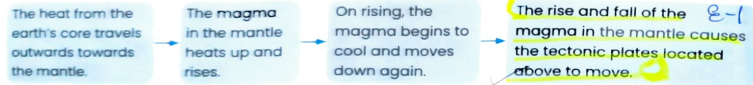
to learn about endogenic forces.



Fig. 2.2: Major and minor tectonic plates

Can you tell whether the ground on which you stand is still or moving? It seems still, does it not? However, the land that we stand on is actually moving! This is because the tectonic plates are constantly moving. But why do the plates move? The answer to this lies in the layers of the earth.

We have learned in Chapter 1 that the earth is divided into layers, namely the crust, the mantle, the inner core, and the outer core. The core, being the hottest layer, emits heat that travels outwards, causing certain changes that lead to the movement of the tectonic plates. Let us see how this happens.

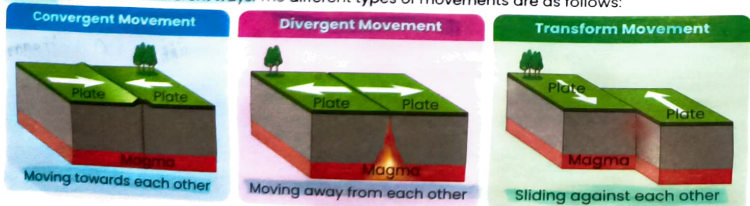


However, this movement of the tectonic plates is so slow that we do not notice it. Most tectonic plates move at a rate of about 2 cm per year, which is slower than the rate at which our fingernails grow!



Fig. 2.3: The rise and fall of magma causes the plates to move.

The tectonic plates move in different directions. When these plates move, they interact with each other at their boundaries in different ways. The different types of movements are as follows:



These three types of tectonic movements have different effects on the earth's surface. These effects are as follows:



Formation of mountains



Volcanoes



Earthquakes

Formation of Mountains

When two tectonic plates meet or collide with each other, sometimes, one of the plates is pushed down and the other plate is pushed up. This leads to the formation of mountains.

For example, the Himalayas were formed when the Indian Plate moved north and collided with the Eurasian Plate. As the plates continue to move, the height of the Himalayas continues to increase every year, even today.

Plate being pushed upwards to form mountains

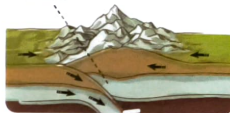


Fig. 2.4: Formation of a mountain

Volcanoes

In the Andaman Sea, there lies a small island called Barren Island. In the middle of this island, there is a cone-shaped mountain with an opening on top. In 2017, fiery hot molten lava began to erupt from the opening. The lava slowly made its way down the slope, burning everything in its path.

The passage above describes a volcanic eruption. But what is a volcano? Why does it erupt? Let us find out.



Fig. 2.5: A volcano

A volcano is an opening in the earth's crust through which molten lava suddenly erupts. Let us look at what happens inside a volcano.

KEYWORDS

erupt: get thrown out suddenly



Fig. 2.6: Structure of a volcano

1. Magma collects below the surface of the earth in an area known as the magma chamber.
2. When the pressure in the magma chamber becomes extremely high, the magma erupts through a pipe-like opening in the earth's crust known as a vent. The magma that comes out of a volcano is called lava.
3. A volcanic eruption throws out lava, hot gases, and ash.

Volcanic eruptions can happen due to different reasons

- When **plates converge**, the plate that is pushed under the other melts due to heat and the magma below the plates rises
- When **plates diverge**, the magma rises through the gap between the two plates
- In some cases, the magma heats up and rises in certain hot areas inside the earth known as **hotspots**.

Types of Volcanoes

Based on their activity, volcanoes can be divided into the following three types:

Active Volcanoes

- Volcanoes that erupt frequently
- Example: Mt Fuji in Japan and Barren Island in India



Dormant Volcanoes

- Volcanoes that have not erupted in a long time but may erupt in the future
- Example: Mt Kilimanjaro in Tanzania



Extinct Volcanoes

- Volcanoes that have not erupted for thousands of years and are unlikely to erupt again
- Example: Mt Popa in Myanmar



Earthquakes



I remember that day clearly. It was around 830 in the morning on 26 January 2001. I was at home with my family in Ahmedabad. Suddenly, the ground started to shake. I froze, not knowing what was happening. My father shouted, 'Earthquake!' We made us get under the dining table. Around us, glass from the windows was falling and shattering. I was terrified! The shaking continued for several minutes. When it finally stopped, we slowly made our way out. Our house was okay, but the building at the end of the street had collapsed! Many people had lost their homes and lives. Later, we found out that the earthquake had been much worse in Bhuj.

The passage above describes a victim's experience of the earthquake that struck Gujarat in 2001. It caused a lot of death and destruction. Why did it happen? What can we do to stay safe during such an event? Let us find out.

Earthquakes are the sudden shaking of the earth, caused due to the movement of tectonic plates. They can occur when two plates slide against each other (transform movement) or move towards each other and collide (convergent movement).

The movement of the two plates releases large waves of energy known as **seismic waves**. These waves travel in all directions, similar to ripples spreading in water when a stone is thrown into it. As the waves travel, the areas they touch begin to shake, causing earthquakes.



to learn about earthquakes.

ACTIVITY 2

Make your own volcano!

- Step 1 First, place an empty plastic bottle in a mound of sand.
- Step 2 Use a funnel to add some baking soda to the bottle.
- Step 3 Mix some food colouring and vinegar together and pour this mixture into the bottle. Watch your volcano erupt!

The place where an earthquake originates inside the earth is known as the **focus**.

The point directly above the focus on the earth's surface is known as the **epicentre**.

Seismic waves are strongest near the epicentre.

Seismic waves become less strong as they move away from the epicentre.

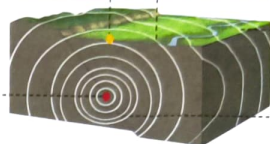


Fig. 2.7: Origin of an earthquake

Because the seismic waves are strongest at the epicentre, places near the epicentre experience the maximum impact. The impact is less in places that are farther away. Based on the above story, can you guess which place (Ahmedabad or Bhuj) was closer to the epicentre of the 2001 Gujarat earthquake?

Not all earthquakes have the same intensity. Some earthquakes cause light shaking, while others can cause a lot of destruction. Scientists measure the **intensity or magnitude of an earthquake** using a number scale known as the **Richter scale**. Earthquakes that measure high on the Richter scale can cause a lot of death and destruction. The earthquake that struck Gujarat in 2001 had a magnitude of 7.7 on the Richter scale. How destructive do you think it was? Look at the scale in Fig. 2.8 to find out.



Fig. 2.8: The Richter scale showing the magnitude of earthquakes

Earthquakes can strike without warning. What can we do to prepare for and stay safe during earthquakes?

Safety Tips to Follow during an Earthquake

- If you are indoors, drop down, take cover under a table, and remain there till the shaking stops.
- If you are outside, move away from tall buildings and get to an open area. Stay away from power lines.
- If you are in a moving vehicle, stop the vehicle and wait.

THINK

Read online sources on how to stay safe during an earthquake. Think about which source you would trust and why.

COLLABORATE

In groups, plan and design a poster to help spread awareness in your school on how to prepare for and stay safe during an earthquake.

DO YOU KNOW?

Countries like Indonesia and Japan experience the most number of earthquakes in the world. This is because they lie on the 'Ring of Fire' where several tectonic plates meet.

The map in Fig. 2.9 shows the distribution of volcanoes and earthquakes across the world. Observe this map and the map in Fig. 2.2 showing tectonic plates. Do you see any similarities between them?

Since volcanoes and earthquakes are caused due to the movement of tectonic plates, most of them occur along the boundaries of the plates. In fact, the boundary of the Pacific Plate experiences almost 75% of the earth's volcanic eruptions and 90% of the earthquakes. Hence, it is known as the Ring of Fire.

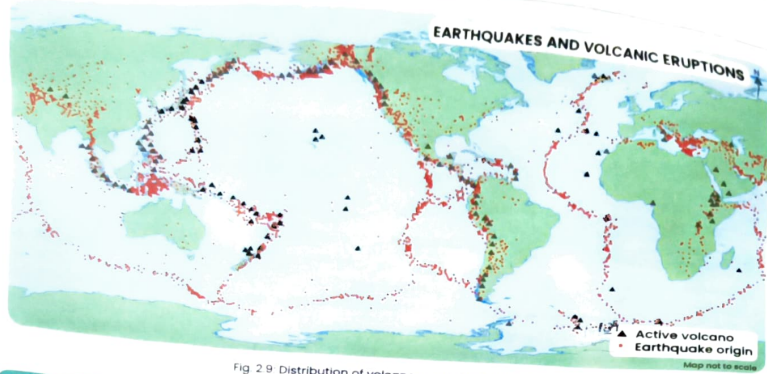


Fig. 2.9: Distribution of volcanoes and earthquakes

2.2 EXOGENIC FORCES

Have you ever seen rocks rolling down the side of a hill? What about winds uprooting trees and blocking roads? We have learned so far about forces inside the earth, but what about forces acting on its surface? Forces that act above the surface of the earth are called **exogenic forces** or external forces. These include the actions of rivers, sea waves, wind, and ice.

These exogenic forces change the landforms around us, breaking them down or building them up. How is it possible for rivers, sea waves, wind, and ice to change strong landforms such as plateaus and mountains? This happens through processes known as weathering, erosion, and deposition.



KEYWORDS

sediment, sand, stones, or mud carried by water or wind and left at the bottom

Processes That Change Landforms

✓ **1. Weathering:** It is the process by which rocks are broken down into smaller pieces.



✓ **2. Erosion:** It is the process by which the broken rock particles are carried away.



✓ **3. Deposition:** It is the process by which eroded sediment is left at a particular place.



These three processes are happening all the time due to the exogenic forces of rivers, sea waves, wind, and ice. Over time, landforms undergo changes through these processes.

Action of Rivers

Rahim lives in a small village in the Himalayas. Every day, he walks along the Ganga river as it rushes past him at high speed. His friend, Aman, lives near the ghats of Varanasi. There, he sees the same Ganga river flow past him. But the river is wide here and flows slowly.

Why is it that Rahim and Aman see the same river in different forms in different places? The answer to this question lies in understanding the journey of a river from its origin until it meets the sea.

A **river** is a natural and continuous stream of water that flows across land to join the sea. At different points during its course, the river weathers, erodes, and deposits the land around it.

- 1 In its upper course, a river usually originates from a mountain. Here, it is fast-flowing, eroding rock as it flows. *if joins valleys and waterfalls*
- 2 In its middle course, the river is joined by **tributaries**, making it larger. Erosion and deposition happen at this stage. *that leads to river flow of sand and silt*
- 3 In its lower course, the river begins to slow down. It deposits many of the sediments it is carrying. *as a result of slow flow*

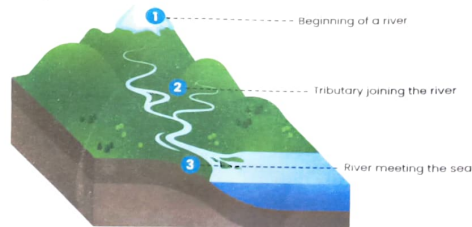


Fig. 2.10: Course of a river

ACTIVITY 3

Do you want to see how rivers erode the land? Let us try an activity

- Put some mud inside a box on one side. Run a finger through the mud like a river.
- Pour water on the side with mud. As the water flows down like a river, observe what happens to the mud.





Meanders are bends in a river as it flows slowly through plains. They are formed by erosion and deposition in the middle course of the river.



Waterfalls are formed in the upper course when rivers erode rocks vertically. This results in the river falling from a height with force.

For example, Jog Falls and Dudhsagar Falls.



Valleys are formed in the upper course as rivers cut through and erode the sides of mountains, creating low-lying areas in between.

For example, the Chenab Valley and Parvati Valley.

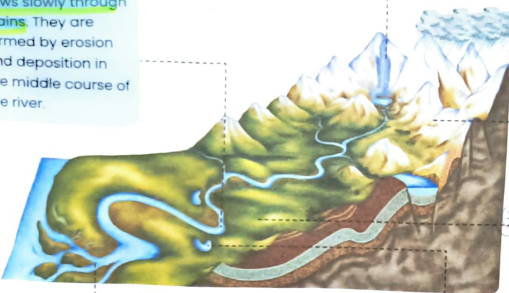


Fig. 2.11 Landforms created by the action of rivers



Deltas are triangular landforms formed in the lower course of a river when it joins the sea. The slow river deposits its sediments and splits into smaller streams, thus creating a delta.

For example, the Sundarbans Delta and Kaveri River Delta.



Oxbow lakes are formed when a river in its middle course deposits so much sediment along its banks that it cuts off a bend in the meander.

For example, Kanwar Lake and Vynthala Lake.



Floodplains are flat, fertile plains formed when a river floods the land and deposits its sediments. They are formed in the lower course of a river.

For example, the Indo-Gangetic Plain and Brahmaputra Plain.

CONNECT TO LIFE

Go through the different landforms formed by the rivers and answer the questions given below.

- Which of these landforms have you seen?
- Which of these landforms are you eager to know more about? Why?

Action of Sea Waves

Have you ever dug your foot into the sand on a beach? How did the soft sand get there? The answer to this lies in understanding the power of sea waves. Sea waves are a strong force, and they create a variety of landforms through erosion and deposition. Let us learn about some of these landforms.

Sea waves continuously strike the rocks along the shore and crack them. These cracks widen and create hollow portions called **sea caves**.

Over time, the waves erode the inside of the caves until only the roof is left. These landforms are called **sea arches**.

Over time, the waves also erode the roof of the sea arches until only the walls are left. These are called **sea stacks**.



Fig. 2.12 Landforms created by the action of waves

When large waves crash against a rocky shore, they slowly erode the softer rocks and leave hard, vertical slopes. These are called **sea cliffs**.

Sea waves deposit sand and other sediments along their shores. The landform created by these deposits over time is called a **beach**.

ACTIVITY 4

Let us do an experiment to demonstrate the action of waves.

- Fill one half of a box tightly with sand as shown in the image. Pour water into the empty half of the box.
- Push the water towards the sand to resemble the motion of waves. Observe what happens to the sand.



Action of Wind

Just like water, wind is a powerful force. It not only blows away sand but also erodes large rocks. Let us learn about some of the landforms created by wind.

Mushroom rocks are formed when low-blowing winds erode the softer lower parts of rocks.



Sand dunes are small hills of sand formed when the wind carries and deposits grains of sand in another location.



Loess is a yellowish-brown, mineral-rich deposit of fine sand (silt). It is formed when silt is carried over a long distance by wind.



Action of Ice



We know that ice is the frozen form of water. But did you know that there are **slow-moving rivers of ice called glaciers**? As glaciers move, they weather and erode the rocks around them and deposit sediments, creating different landforms. Let us learn about some of these landforms.

Glaciers weather and erode landscapes and create hollow spaces that get filled with ice. When the ice melts, it becomes a **glacial lake**.



As glaciers move, they create deep, hollow spaces in the shape of an armchair. These landforms are known as **cirques**.

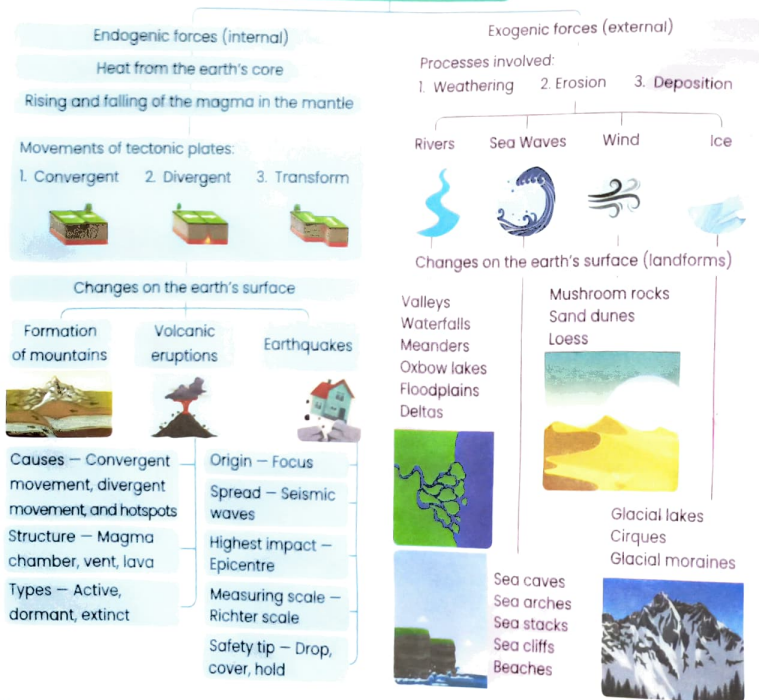


Glaciers deposit rocks and sediments as they flow, forming a landform called **glacial moraine**.



LET'S REVISE

Forces Acting on the Earth



EXERCISES

A. Fill in the blanks.

1. The individual pieces of the lithosphere are known as tectonic plates or plates.
2. An earthquake is the sudden shaking of the earth, caused due to the movement of the tectonic plates.
3. Mt Fuji is an active volcano as it erupts frequently.
4. Meanders are formed by a river in its middle course when the deposition of sediments cuts off a bend in the meander.
5. The landform created due to the deposition of sand and other sediments along the shores by the sea waves is called a beach.
6. Glaciers deposit rocks and sediments as they flow, forming glacial moraine.

B. Write True or False. Rewrite the false statements correctly in your notebook.

1. Most tectonic plates move at a rate of 2 km per year. True
2. The focus is the place inside the earth where an earthquake originates. True
3. Weathering happens when rocks are carried away. False (Weathering is the breaking down of rocks.)
4. Deposition leaves sediments at a particular place. True
5. Waterfalls are formed in the lower course when rivers erode rocks vertically. True

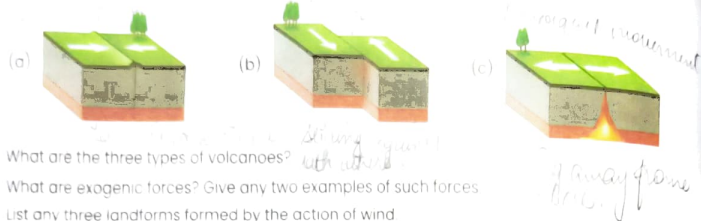
C. Choose the correct answer.

1. Which of the following is used to measure the intensity of an earthquake?
 - (a) Epicentre
 - (b) Seismic wave
 - (c) Richter scale
 - (d) Focus
2. Which of the following is/are thrown out during a volcanic eruption?
 - (a) Lava
 - (b) Ash
 - (c) Hot gases
 - (d) All of the above
3. Which of the following landforms is formed in the lower course of a river?
 - (a) Waterfalls
 - (b) Valleys
 - (c) Delta
 - (d) Stacks
4. Which of the following are small hills of sand formed when the wind carries and deposits grains of sand in another location?
 - (a) Mushroom rocks
 - (b) Moraines
 - (c) Sand dunes
 - (d) Loesses
5. Which of the following statements correctly states how cirques are formed?
 - (a) Formed when glaciers deposit rocks and sediments as they flow
 - (b) Formed when sea waves deposit rocks and sediments as they hit the shores
 - (c) Formed when glaciers create hollow spaces that get filled with ice
 - (d) Formed when glaciers create deep, hollow spaces in the shape of an armchair

D. Answer the following questions in brief:

1. Define endogenic forces.

2 Identify and describe the tectonic plate movements shown below



- What are the three types of volcanoes?
- What are exogenic forces? Give any two examples of such forces.
- List any three landforms formed by the action of wind.

E. Answer the following questions in detail:

- What are tectonic plates? Why do they move?
- What is a volcano? Explain how it erupts.
- Describe the course of a river.
- Copy the following table in your notebook and fill it out for the different landforms formed by the action of rivers and waves. One has been done as an example.

Exogenic Force	Landform	Description	Drawing
Action of rivers	Valley	Valleys are low-lying areas between mountains that have been eroded by rivers.	

- Explain how glaciers create landforms.

F. HOTS — Think and answer

- Mount Everest is located in the Himalayas at the boundary of the Indian and Eurasian tectonic plates. Use this information to answer the following questions:
 - How was Mount Everest formed?
 - Why is its height increasing every year?

G. Competency-based questions

- Ria and her family are moving to Japan because her father got a new job. Ria was curious about Japan and found out that it is located on the Ring of Fire. This made her feel worried.
 - Why do you think Ria is worried after learning that Japan is on the Ring of Fire?
 - What steps can Ria and her family take to stay safe in Japan?
- If you are travelling by car from Amritsar in Punjab to Shimla in Himachal Pradesh, what are the different landforms formed by rivers that you might come across on your journey?

REFLECTIONS

- Which of the two forces — endogenic or exogenic — did you find more interesting? Why?



India and the World: Land and the People

3. Atmosphere



IN THIS CHAPTER, YOU WILL ...

- Learn about the composition and structure of the atmosphere
- Understand the greenhouse effect and its impact on the earth.
- Differentiate between weather and climate.
- Understand how changes in temperature and pressure affect our lives.
- Understand how winds blow and how rainfall is caused.

CONNECTIONS TO OUR LIFE

- The air we breathe is made up of many things.
- Cloudy, windy, and rainy days are caused by changes in the atmosphere.
- A hill station is usually colder than a place located in the plains due to differences in air pressure.

BIG IDEA

The atmosphere is important for life on the earth. Many events, such as rainfall and wind, are caused due to constant changes that take place in the atmosphere.